



Affirmed Networks Terms and Acronyms

Updated: October 2019

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Preface

This document lists and describes terms and acronyms used in the Affirmed Networks documentation.

Audience

This document is intended for network operators (sometimes referred to as Users or Operators in this document), installation staff, and other qualified service personnel who want to install, provision, monitor and learn about the Affirmed Networks system and associated features and functions. The audience should also be familiar with Long Term Evolution (LTE), and 3G or 4G networks. It may also be helpful to have a basic understanding of the HTTP protocol, TCP/IP stack, and the use of MMS services.

How to Obtain Product Documentation

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Terms and Acronyms

The following terms and acronyms may appear in the Affirmed Networks documentation.

Term or Acronym	Description
10GBASE-BX4	Common name of the standard that defines 10 gigabyte Ethernet data link over eight pairs of copper, four pairs for transmit and four pairs for receive.
1Q	See DOT1Q.
1xRTT	CDMA2000 Radio Transmission Technology
2G	Second generation wireless mobile telecommunications technology
3DES	Triple DES Technique where DES is performed multiple times to increase security level.
3G	Third generation wireless mobile telecommunications technology
3GPP	Third Generation Partnership Project – produce globally applicable technical specifications for 3G mobile systems based on evolved GSM.
4G	Fourth Generation wireless mobile telecommunications technology
5G	Fifth Generation wireless mobile telecommunications technology
5G NR	Fifth Generation New Radio network
5G-NSA	Fifth Generation Non-Standalone network
5G-NSA NR	Fifth Generation Non-Standalone network New Radio solution
5G SA	Fifth Generation Standalone
A/AAAA	A – refers to the IPv4 address space. AAAA refers to the quad-A record – refers to the IPv6 address space.
AAA	Authentication, Authorization, and Accounting – key functions to intelligently control access, enforce policies, audit usage, and provide the information necessary to perform billing for services. AA-Answer
AAFA	Active Flow Analytics

Term or Acronym	Description
AAR	AA-Request
ABNF	Augmented Backus-Naur Form
ABQP	Aggregate BSS QoS Profile
ABR	Area Border Router
ACL	Access Control List
ACR	Anonymous Customer Reference – an encrypted MSISDN.
ACS	Access Controller Server
ADC	Analog-to-Digital Conversion
ADMF	Administration Function Administration Function for Legal Intercept
AdvancedTCA	Advanced Telecommunications Computing Architecture
AE	Application Entity
AES	Advanced Encryption Standard A symmetric key encryption technique.
AF	Access Function
AFC	Active Failover Cluster – consists of nodes that are all active and ready to provide services channeled through a load-balancer/router.
AGW	Access Gateway
AIS	Application Interface Specification
AJP	Apache JServ Protocol – packet-based protocol that allows HTTP Web Server to communicate with Servlet Engine instances.
AKA	Authentication and Key Agreement
ALG	Application Layer Gateway
AMBR	Aggregated Maximum Bit Rate
AMC	Advanced Mezzanine Card
AN-SCEF	Affirmed Networks Service Capability Exposure Function VNF that implements the 3GPP SCEF+MTC-IWF functions specified in 3GPP 23.682[1].
ANSI	American National Standards Institute
AOC	Advice of Charge
APCO	Additional Protocol Configuration Option
API	Application Programming Interface

Term or Acronym	Description
APN	Access Point Name – gateway name between a GPRS, 3G, or 4G mobile network and the public Internet. A mobile device making a data connection must be configured with an APN to present to the carrier. APN identifies the packet data network (PDN) with which a mobile data user wants to communicate.
APN-AMBR	Access Point Name - Aggregated Maximum Bit Rate
APN-NI	APN Network Identifier
APN-OI	APN Operator Identifier
APS	Automatic Protection Switching
ARP	Address Resolution Protocol Allocation and Retention Policy Allocation and Retention Priority
AS	Application Server
ASAP	Affirmed Service Automation Platform
ASM	Advanced Services Module – a VM instance that runs the tasks needed for providing advanced services like HTTP Proxy, Cache, WAP GW, etc.
ASN.1	Abstract Syntax Notation One
ASP	Application Server Process, as defined by RFC 4666
ASR	Abort-Session-Request
Asset	Refers to vBuilder vendor objects, vendor components, service components, and service models.
Association Service Component	Used by a service model to define the relationship between a “From” service component and a “To” service component in an association. It can be a regular service component that represents associated objects for specific devices based on their schema, or it can be an ‘empty’ service component that only represents the relationship between the service components but not the associated objects on the vendor object instance/device instance.
ATCA	See AdvancedTCA.
ATM	Asynchronous Transfer Mode
AUSF	Authenticated Server Function
AVOD	Advertisement Video on Demand
AVP	Attribute Value Pair
AWG	American Wire Gauge
AWT	Average Working Time
B-Frame	Bidirectionally-predicted Frame – opens up the ability for the encoder to send frames in an arbitrary order, so the decoder may need to decode multiple frame before displaying them. Frames can be predicted both backward time and forward time (forward referenced frames need to be sent out of order).

Term or Acronym	Description
BBERF	Bearer Binding and Event Reporting Function
BBS	Basic Blade Service
BCD	Binary-coded decimal – a family of representations of numerals, uppercase Latin letters, and some special and control characters as six-bit character codes.
BER	Basic Encoding Rules
BFD	Bidirectional Flow Detection Bidirectional Forwarding Detection
BGP	Border Gateway Protocol
BH	Busy Hour
BIX	Byte Information Exchange
BS	Base Station Billing System
BSC	Base Station Controller – the BSC is in charge of the allocation and release of radio channels and handover management.
BSN	Broadband Service Node
BSS	Business Support System Base Station System or Subsystem
BSSAP	Base Station System Application Part
BSSGP	Base Station System GPRS Protocol
BT	Block Transfer
BVC	BSSGP Virtual Connection – an end-to-end virtual communications path between remote network service user entities.
BVCI	BSSGP Virtual Connection Identifier
C-IoT CIoT	Cellular Internet of Things
C-Plane	Control Plane – administration functions outside the transfer of user data packets associated with functions such as setting up and tearing down of connections, establishing identities, and authorizing users.
C-SGN	Cellular Serving Gateway Node Cellular Internet of Things (CIoT) Serving Gateway Node
CALEA	Communications Assistance for Law Enforcement Act
CallP	Call Processing, a type of Virtual Machine on the MCC MME
CAMEL	Customized Application for Mobile Network Enhanced Logic

Term or Acronym	Description
CAP	CAMEL Application Part Common Administration Part
Capex	Capital Expenditure
CAR	Common Alarm Record
CATT	Common Automated Test Tool
CBC	Cell Broadcasting Center
CBR	Constant Bit Rate – useful for streaming multimedia content on limited capacity channels, as it takes advantage of all available capacity.
CC	Content of Communications
CCA	Credit-Control-Answer message
CCLID	CC Link Identifier
CCM	Centralized Control Module – a VM instance that runs the centralized tasks needed for call-control functionality and routing protocols.
CCR	Credit-Control-Request message
CDF	Charging Data Function
CDMA	Code Division Multiple Access
CDN	Content Delivery Network
CDP	Common Delivery Part
CDR	Charging Data Record – a formatted collection of information about a chargeable event (for example, time of call set-up, duration of the call, amount of data transferred, and so on) for use in billing and accounting. For each party to be charged for parts of or all charges of a chargeable event, a separate CDR is generated. Multiple CDRs may be generated for a single chargeable event, for example, because of its long duration, or because more than one charged party is to be charged. Also referred to as Call Data Record or Call Detail Record.
CFC	Cold Failover Cluster Consists of one primary node that serves requests and one or more failover or standby nodes. At any given time, only one node in the cluster is alive and manages all system resources.
CFDS	Customer Feature Delivery System
CG	Charging Gateway – the device on the GSM GPRS or UMTS network that collects and maintains Call Detail Records (CDRs) for subscriber PDP contexts. Also referred to as a Charging Gateway Function (CGF).
CGF	Charging Gateway Function – see CG.
CGI	Cell Global Identity
CHAP	Challenge Handshake Authentication Protocol

Term or Acronym	Description
CIA	Configuration Information Answer
CID	Communication Identifier
CII	Communication Identifying Information
CIN	Communication Identity Number
CIQ	Customer Information Questionnaire
CIR	Configuration Information Request
CK/IK	Cipher Key/Integrity Key
CLI	Command Line Interface
CM	Configuration Management
CMA	Connection Management Answer
CMAS	Commercial Mobile Alert System messages
CMR	Connection Management Request
CN	Compute Node Core Network
CNA	Cloud Native Architecture
CoA	Change of Authorization
Coalescing	A grouping of parameters that share the same base type object class and are configured with the same values. This feature simplifies configuration and reduces data entry errors. The value supplied to a coalesced parameter is propagated to all child parameters.
codec	Device or computer program capable of encoding or decoding a digital data stream or signal.
Compression Gain	Ratio of bytes-in to bytes-out.
CoS	Classes of Service
COTS	Commercial Off The Shelf
CP	Control Processor – control processors that manage and control applications supporting both system and network functions. Control Plane – see C-Plane.
CPA	Certification Path Advertisement
CPE	Control Plane Engine
CPF	Control Plane Function
CPM	Control Plane Module
CPR	Common Performance Record

Term or Acronym	Description
CPS	Certification Path Solicitation
CPU	Central Processing Unit
CRBN	Charging-Rule-Base-Name
CRC	Cyclic Redundancy Check – the monitoring of a digital bit stream to detect deviations from the expected bit patterns.
CRR	Call Release Reason
CS	Circuit Switched Corporate Standards
CSCF	Call State Control Function
CSFB	Circuit Switched Fallback
CSG	Closed Subscriber Group
CSGID	Closed Subscriber Group Identity
CSI	CAMEL Subscription Information
CSL	Call Summary Log
CSM	Control/Content Services Module – a VM instance that runs all services provided by CPM and ASM.
CSR	Create Session Request
CSV	Comma Separated Value
CUPS	Control and User Plane Separation
DASH	Dynamic Adaptive Streaming over HTTP
Data Hiding	Filtering out unused (optional) parameters and parameters whose values have already been set. Data hiding is accomplished by applying the Hard Code or Ignore decoration to a parameter.
DB	Database Database files (such as control files, redo-log files, and data files) and a set of processes that manage them along with various system mechanisms (such as shared memory and semaphores).
DC	Direct Current Diameter Client
DCCA	Diameter Credit-Control Application
DCM	Distributed Control Module
DCNR	Dual Connectivity with New Radio
DDR	Double Data Rate
DEA	Diameter-EAP-Answer – a command sent by the Diameter server to the client.

Term or Acronym	Description
DER	Diameter-EAP-Request – a command sent by a Diameter client to a Diameter server that conveys an EAP-Response from the EAP client.
DES	Data Encryption Standard A symmetric key encryption technique.
Device Instance	An instance of an actual device, for example, a device, application, tool, etc. that is registered in and known by vTransactor.
Device Type	A generic term used to define one or more vendor devices, network elements (NE), or applications, such as, Affirmed MCC, MCC EMS, and Cisco PCRF.
DF	Delivery Function Delivery Function for Legal Intercept
DFA	Data Flow Analysis
DG	Device Gateway
DHCP	Dynamic Host Configuration Protocol
DIMM	Dual In-line Memory Module
DMZ	Demilitarized Zone – refers to one or more computers that are outside the firewall in a networking context. These computers/servers broker requests from outside (Internet) as well as the rest of the local area network.
DNS	Domain Name System – a system that stores information associated with domain names in a distributed database on networks, such as the Internet. Part of the IP protocol stack, this allows host name to IP address mapping within an Internet environment. Domain Name Service
DOIC	Diameter Overload Indication Conveyance
DoNAS	Data over Non Access Stratum
DOT1Q	IEEE 802.1q – the networking standard that supports virtual LANs (VLANs) on an Ethernet network.
Downlink Traffic	HSDPA traffic traveling from the RNC to the UE.
DP	Discard Priority
DPDK	Data Plane Development Kit – a set of data plane libraries and NIC drivers for fast packet processing.
DPE	Data Plane Engine
DPFW	Data Plane Framework
DPI	Deep Packet Inspection
DPR	Disconnect Peer Request
DRA	Delivery Report Answer DHCP Relay Agent Diameter Routing Agent

Term or Acronym	Description
DRA/SLF	Diameter Routing Agent/Subscriber Location Function
DRM	Digital Rights Management
DRMP	Diameter Message Priority
DRR	Delivery Report Request
DRT	Data Record Transfer
DSCP	Differentiated Services Code Point – a field in the IPv4/6 headers that describes the desired QoS of a packet.
DST	Daylight Savings Time
DTA	Device Trigger Answer
DTR	Device Trigger Request
DUPMS	Distributed UDP Path Management System
DUT	Device Under Test
DVL	Data Volume Limit
DVLT	Data Volume Limit Trigger
DWA	Device-Watchdog-Answer
DWR	Device-Watchdog-Request
E-RAB	E-UTRAN Radio Access Bearer
E-SMLC	Evolved Serving Mobile Location Center
E-UTRAN	Evolved UMTS Terrestrial Radio Access Network
E-UTRAN	Evolved UMTS Terrestrial Radio Access Network
E-UTRAN Node B	Evolved Node B
eAN	Evolved Access Node for CDMA HRPD network
EAP	Extensible Authentication Protocol
EAP-AKA	Extensible Authentication Protocol - Authentication and Key Agreement
EAP-GTC	Extensible Authentication Protocol - Generic Token Card
ECC	Enterprise Cloud Communications
ECC	Embedded Communications Computing
ECDSA	Elliptic Curve Digital Signature Algorithm – offers a variant of the Digital Signature Algorithm (DSA) which uses elliptic curve cryptography.
ECGI	E-UTRAN Cell Global Identity

Term or Acronym	Description
ECM	EPC Connection Management
EDP	Event Detection Point
EDR	Event Data Record
EEPROM	Electrically Erasable Programmable Read-only Memory
EGP	Exterior Gateway Protocol
EGS	Entitlement Generation Server
EIR	Equipment Identity Register
EJB	Enterprise JavaBeans
ELP	EPC LCS Protocol
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EMM	EPS Mobility Management
EMS	Element Management System Enhanced Messaging Service
EN	European Norm
eNB eNodeB	Evolved Node B
EP	Emission Priority
EPC	Evolved Packet Core – provides a converged voice and data-networking framework to connect users on a Long-Term Evolution (LTE) network.
EPC	Enhanced or Evolved Packet Core
ePDG	Evolved Packet Data Gateway
EPG	Electronic Program Guide
EPLMN	Equivalent PLMN
EPS	Evolved Packet System
eRANAP	enhanced RANAP (Radio Access Network Application Protocol)
eSCEF	Affirmed Networks enhanced Service Capability Exposure Function product that is a composite VNF (cVNF) that instantiates the DG and the AN-SCEF VNFs to provide a complete API based IOT solution (for transport and other services such as monitoring) for service providers.
ESD	Electrostatic Discharge
ESME	External Short Messaging Entities

Term or Acronym	Description
ESP	Encapsulated Security Payload
ETRAN	Enhanced Terrestrial Radio Access Network
ETSI	European Telecommunication Standards Institute
EV-DO	EV-Data Optimized
EVOD	Exclusive Video on Demand
E/W	East/West traffic
FACH	Forward Access Channel
FCC	Federal Communications Commission
FD	Functional Description
FFS	For Future Study
FGW	Femtocell Gateway, also called small cell gateway
FIFO	First In, First Out
FM	Fault Management
Footprint	Represents the collection of devices or applications (device instances) that are affected when you roll out a snapshot. See also Snapshot (page 30) .
FP	Fastpath Framing Protocol Functional Processor
FPGA	Field Programmable Gate Array
FQDN	Fully Qualified Domain Name
FQPC	Fully Qualified Partial CDR
FRS	Feature Requirements Specification Functional Requirements Specification
FRU	Field Replaceable Unit
FSM	Finite State Machine
FTP	File Transfer Protocol
FTS	Feature Technical Specification
FUBM	Fair Use Bandwidth Management
FUI	Final-Unit-Indication
FVOD	Free Video on Demand

Term or Acronym	Description
Ga	Interface serving the CDRs that are written in the GSN and sent to the charging gateway (CG). The Ga interface uses a GTP-based protocol, with modifications to support CDRs (Called GTP'). The interface is between the SGSN and the CGF.
GB	Gigabyte
Gb	Interface between the base station subsystem BSS and the SGSN. The transmission protocol can be Frame Relay or IP.
Gbps	Gigabits per second
GBR	Guaranteed Bit Rate
GCSNA	Circuit Services Notification Application
Gd	Interface between the SGSN and the SMS Gateway. The Gd interface can use MAP1, MAP2 or MAP3. The Gd interface is between the SGSN and the GSM SMS-GMSC/SMS-IW MSC.
Ge	Interface between the GSN and the service control point (SCP). Uses the CAP protocol. The Ge interface is between the SGSN and the EIR.
GEC	Global Emergency Call
GERAN	GPRS EDGE Radio Access Network
Gf	Interface between the SGSN and the Equipment Identity Register (EIR). It is used for checking the International Mobile Equipment Identity (IMEI) against a list of reported stolen mobile phones.
GGSN	Gateway GPRS Support Node – interface between the GPRS wireless data network and other networks such as the Internet or private networks. It supports the edge routing function of the GPRS network. The GGSN is the node within the GPRS core network that is responsible for inter-working with external PDNs, such as the Internet.
GI	Geographical Identity
Gi	1. IP based interface between the GGSN and a public data network (PDN), either directly to the Internet or through a WAP gateway. 2. An external interface between the GGSN and another packet data network.
Gi-LAN	A 3GPP term for a network upstream from the PGW. In Gi-LAN, value-added services are executed including TDF, firewall, Carrier-grade NAT, TCP optimization, traffic optimization, and URL filtering.
GiGW	Gi Gateway – see Gi.
Gmb	Interface between the GGSN and the Broadcast-Multicast Service Center (BM-SC) that is used for controlling MBMS bearers.
GMLC	Gateway Mobile Location Center
GMM	GPRS Mobility Management
GMM context	GPRS Mobility Management – the GMM context is the information sets held at the MS and the SGSN. It contains mobility-related information such as the Mobility Management state as well as location information, IMSI, and negotiated timer values.

Term or Acronym	Description
Gn	IP-based interface between the SGSN and other SGSNs and (internal) GGSNs. The interface uses the GTP Protocol. DNS also shares this interface.
GNP	Generic Network Product – a class of network products that all implement a common set of 3GPP-defined functionalities for that particular network product.
Gp	IP-based interface between the internal SGSN and external GGSNs. The interface uses the GTP Protocol.
GPP	3rd Generation Partner Project
GPRS	General Packet Radio Service – a packet radio access technique based on GSM radio to transfer data in an efficient manner, optimizing the use of network resources. The GPRS re-uses existing GSM radio technology.
GPS	Global Product Support
Gr	Interface between the SGSN and the HLR. Messages going through this interface use the MAP3 protocol.
GRE	Generic Routing Encapsulation
Gs	Interface between the SGSN and the MSC (VLR). The interface uses the BSSAP+ protocol. The Gs interface allows paging and station availability when it performs data transfer. The Gs interface provides mobile stations the ability to communicate with the MSC/VLR for circuit switch services while they are attached to the packet network.
GSM	Global System for Mobile communications – a technical standard for European mobile networks.
GSN	GPRS Support Node – constitutes the interface between the radio system and the fixed networks for packet switched services. The GSN performs all necessary functions in order to handle the packet transmission to and from the mobile stations. The GPRS Support Node can be either SGSN or GGSN.
GSU	Granted Service Units (Container)
GT	Global Title
GTP	GPRS Tunneling Protocol – the transport protocol used between the SGSN and GGSN for both signaling and user data transfer. (C-Plane or U-Plane)
GTP-C	GPRS Tunneling Protocol - C-Plane – the protocol for the control plane handling of signaling between the GSNs within the core network. GPRS Tunnel Protocol-Control is the protocol between the subscriber control and the GGSN used to communicate on behalf of the GMM and SM functions.
GTP-U	GPRS Tunneling Protocol - U-Plane – the protocol used for carrying user data within the GPRS core network and between the Radio Access Network and the core network.
GTPP	GTP Prime Protocol
GTT	Global Title Translation
GUMMEI	Globally Unique Mobility Management Entity Identifier
GUTI	Globally Unique Temporary Identity
GW	Gateway

Term or Acronym	Description
Gx	Off-line policy interface between the GGSN and the charging rules function (CRF). It is used for provisioning the service data flow based on charging rules. The interface uses the diameter protocol.
Gy	On-line charging interface between the GGSN and the online charging system (OCS). The interface uses the diameter protocol (DCCA application).
Gz	Off-line (CDR-based) charging interface between the GSN and the CG. The interface uses GTP'.
HA	High Availability
Hello Packet	In the Open Shortest Path First (OSPF) communications protocol that enables network routers to share information with each other, a Hello packet is a special packet (message) that a router sends out periodically to establish and confirm network adjacency relationships.
Hello Protocol	OSPF creates adjacencies between neighboring routers to exchange routing information. Not every two neighboring routers will become adjacent. The Hello Protocol is responsible for establishing and maintaining neighbor relationships. It also ensures bidirectional communication between neighbors. HELLO packets are sent periodically out all router interfaces. Bidirectional communication is indicated when the router sees itself listed in the neighbor's HELLO packet.
HFN	Hyper Frame Number
HI	Handover Interface
HI1	Handover Interface Port 1 (for Administrative Information)
HI2	Handover Interface Port 2 (for Intercept-related Information)
HI3	Handover Interface Port 3 (for Content of Communication)
HLR	Home Location Register – used to store the subscription record, temporary data, the serving-MSC number and VLR number of a mobile subscriber.
HLS	HTTP Live Streaming
HPLMN	Home PLMN – this represents the operator of an incoming roamer.
HRPD	High Rate Packet Data, also referred to as 1xEV-DO for high speed data over CDMA.
HSS	Home Subscriber Server – the concatenation of the HLR (Home Location Register) and the AuC (Authentication Center). The HLR part of the HSS stores and updates the database containing all subscription information, when necessary.
HTDR	HTTP Transaction Data Record
HTTP	HyperText Transport Protocol
I/O	Input/Output
IAP	Intercept Access Point
ICD	Interface Control Document
ICE	Interception Control Elements
ID	Identifier

Term or Acronym	Description
IDi	Identification:Initiator – an identification payload that allow peers to assert an identity to one another. In the case of the ePDG, the IDi is always the UE.
IDR	Insert Subscriber Data Request
IDr	Identification:Responder – an identification payload that allow peers to assert an identity to one another. In the case of the ePDG, the IDr is always the ePDG.
IE	Information Element
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IKE	Internet Key Exchange – control plane protocol used by IP Security Protocol (IPsec).
IKE	Internet Key Exchange
ILF	IMSI Locator Function
IM	Installation Method
IMC	IPMC Master Controller
IMEI	International Mobile Equipment Identity
IMEISV	International Mobile Equipment Identity of the equipment served.
IMEISV	International Mobile Equipment Identity Software Version
IMS	International Mobile Subscriber IP Multimedia Subsystem IP Multimedia Services
IMSI	International Mobile Subscriber Identity
IN	Intelligent Network
IOM	I/O Module – a VM instance that is responsible for receiving packets into the MCC and sending the packets out.
IOS	Interoperability Specification
IoT	Internet of Things
IP	Internet Protocol – one of the fundamental protocols of the TCP/IP Internet Protocol suite. It defines the IP datagram as the unit of information passed across an Internet and provides the basis for connection-less, best-effort packet delivery service.
IPC	Inter-Process Communication
IPMB	Intelligent Platform Management Bus
IPMC	Intelligent Platform Management Controller Intelligent Peripheral Management Controller

Term or Acronym	Description
IPMI	Intelligent Platform Management Interface
IPsec	IP Security Protocol
IPSP	IP Server Process as defined by RFC 4666. IPSP is essentially the same as ASP. For the SGSN, IPSP is used to denote the function interfaces with the RNC, whereas ASP is used to denote the function interfaces with the core SS7 network.
IRAT	Inter Radio Access Type
IRAU	Inter-SGSN Routing Area Update – a procedure that enables a subscriber to move between cells covered by different SGSNs while maintaining an active PDP context.
IRI	Intercept Related Information
ISC	IPMC Slave Controller
ISD	Insert Subscriber Data
ISM	Integrated Services Module
ISO	International Organization for Standardization
ITU	International Telecommunication Union
Iu	General term for referring to 3GPP interface that links the RNC with the GGSN.
IuCS	3GPP circuit-switched interface that links the RNC with the MSC.
IuPS	3GPP packet-switched interface that links the RNC with the GGSN.
IVOD	Interactive Video on Demand – VOD with interactive features such as play, resume, stop, pause, fast forward, and with the ability to avoid advertisements, purchase goods, and so on.
IWM	Integrated Workflow Module
IWMSC	Interworking Mobile Switching Center
IWS	Interworking System
Java Messaging Service (JMS)	One of the J2EE components involved in relaying messages (point-to-point or publish-subscribe types) between Java objects via TCP/IP in the Java Messaging Service.
JavaScript Object Notation (JSON)	Data exchange format used in the request and response messages through the vTransactor northbound API calls.
JMS	Java Messaging Service – one of the J2EE components involved in relaying messages (point-to-point or publish-subscribe types) between Java objects via TCP/IP.
KASME	Key Access Security Management Entries
KCI	Key Control Indicator
KNI	Kernel NIC Interface
KPI	Key Performance Indicators

Term or Acronym	Description
KS	Knowledge Services
KSI	Key Set Identifier
KVM	Kernel-based Virtual Machine
L2oGRE	Layer 2 over Generic Routing Encapsulation
L2TP	Layer 2 Tunneling Protocol
LAI	Location Area Identity
LAN	Local Area Network
LB	Load Balancer
LB VM	Load Balancer Virtual Machine
LBB	Location Based Billing
LBO	Local Break-Out
LCF	Location Client Function
LCS	Location Services
LCS-AP	LCS Application Protocol
LDAP	Lightweight Directory Access Protocol
LEA	Law Enforcement Agency
LED	Light Emitting Diode
LEMF	Law Enforcement Monitoring Facility
LFU	Least Frequently Used
Lg	Interface between the SGSN and the Gateway Mobile Location Center (GMLC) that is used for location-based services.
LI	Lawful/Legal Intercept
LI ID	Lawful Intercept Identity
LIAP	Lawful Intercept Access Part Lawful Interception Access Protocol
LICP	Legal Interception Common Protocol
LIG	Lawful/Legal Intercept Gateway
LII	Lawful Intercept Identifier
LIID	Lawful Interception Identifier
LIPA	Local IP Access

Term or Acronym	Description
LISP	Lawful Interception Service Processor
LLC	Logical Link Control Logical Link Control layer protocol
LLC layer	Logical Link Control – the upper sublayer of the data link layer of the seven-layer OSI model that provides media-independent functions and the logical connection between the stations within the local area network.
LMS	License Management System
LP	Logical Processor
LPM	Longest Prefix Match
LPP	LTE Positioning Protocol
LPWAN	Low Power Wide Area Network
LRU	Least Recently Used
LSA	Link State Advertisement – a unit of data describing the local state of a router or network. For a router, it includes the state of the router's interfaces and adjacencies. Each link state advertisement is flooded throughout the routing domain. The collected link state advertisements of all routers and networks forms the protocol's link state database.
LSAF	Location Subscriber Authorization Function
LSB	Least Significant Bit
LSBF	Location System Billing Function
LSOF	Location System Operations Function
LSPF	Location Subscriber Privacy Function
LTE	Long Term Evolution
LTMI	Long Term Mobile ID
LTS	Long Term Support
LX	Long Range Optical Transceiver
M-CDR	Mobility CDR
M-TMSI	MME-Temporary Mobile Subscriber Identity
M2M	Machine-to-Machine
M3UA	MTP Level 3 (MTP3) User Adaptation Layer
MAC	Medium Access Control

Term or Acronym	Description
MAP	Mobile Application Part – the MAP is the software protocol layer that allows information about a roaming mobile-subscriber (MS) to flow between functional entities in the Public Land Mobile Network (PLMN).
MB	Megabyte
MBMS	Multimedia Broadcast-Multicast Service
Mbps	Megabits per second
MBR	Maximum Bit Rate
MBSFN	MBMS over a Single Frequency Network
MCA	Management Configuration Adapter
MCC	Mobile Content Cloud Mobile Country Code
MCD	Messaging Channel Database
MCE	Multi-cell/Multicast Coordination Entity
MCG	Master Cell Group
MCM	Management Configuration Module
MD	Mediation Device
ME	Mobile Entity Managed Element Mobile Equipment
MeNB	Master eNB
MF	Mediation Function
MH-BGP	Multi-homing Border Gateway Protocol
MiTM	Man-in-the-middle-attack – when an attacker secretly relays and possibly alters the communication between two parties who believe they are directly communicating with each other.
MLT	Multi Link Trunk
MM	Mobility Management
MMC	Mezzanine Management Controller
MME	Mobility Management Entity
MMS	Multimedia Messaging Service
MMSC	Multimedia Messaging Service Center
MNC	Mobile Network Code

Term or Acronym	Description
MNO	Mobile Network Operator
MO	Managed Object Mobile Originated
MO-LR	Mobile Originated Location Request
MOBIKE	IKEv2 Mobility and Multi-homing Protocol – for more information, see RFC 4555.
MONTE	Monitoring Enhancements
MPC	Modular Port Concentrators
MPLS	Multi-protocol Label Switching
MPU	Main Processing Unit
MRA	Multi-protocol Routing Agent
MRM	Management and Resource Module
MS	Mobile Station – subscriber equipment that provides access into the Public Land Mobile Network (PLMN).
MSC	Mobile Switching Center – provides switching and call control of GSM basic and supplementary services.
MSC/VLR	Mobile Switching Center/Visitor Location Register
MSCC	Multiple Services Credit Control
MSF	Mediation Service Function
MSISDN	Mobile Subscriber Integrated Services Digital Network-Number – a number string used to uniquely identify a mobile telephone subscriber in a GSM or a UMTS mobile network.
MSM	Mobile Service Module
MSS	Maximum Segment Size Multi-service Switch
MT	mobile terminated
MT TDA	Mobile Terminated Data-Answer
MT TDR	Mobile Terminated Data Request
MT-LR	Mobile Terminated Location Request
MTC	Machine-Type Communications
MTC-IWF	Machine Type Communications-Interworking Function
MTU	Maximum Transmission Unit
MVNE	Mobile Virtual Network Enabler

Term or Acronym	Description
MVNO	Mobile Virtual Network Operator
MZ	Militarized Zone – a local area network or a sub network behind a firewall that is not directly accessible by any computer on the Internet.
NA	Not Applicable Neighbor Advertisement
NACC	Network Assisted Cell Change
NAI	Network Access Identifier Nature of Address Indicator
NAPT	Network Address Port Translator
NAPTR	Naming Authority Pointer Resource Record is defined by the IETF for Dynamic Delegation Discovery System (DDDS). It is defined in RFC 3401, 3402, 3403 and 3404.
NAS	Non Access Stratum Network Attached Storage
NAT	Network Address Translation
NB	narrowband
NB-IoT NB-IoT	Narrow Band – Internet of Things (aka Narrowband-IoT) – an LPWAN radio technology standard that has been developed to enable a wide range of devices and services to be connected using cellular telecommunications bands.
NBA	Node Backbone Area
NBR	NAT Binding Records
ND	Neighbor Discovery
NDP	Neighbor Discovery Protocol
NE	Network Element
NE-RAMS	Network Element Reliability, Availability, Maintainability, and Survivability
NEAF	Non-EPS Alert Flag
NEBS	Network Equipment Building System
NFA	Network Flow Analysis
NFV	Network Functions Virtualization – allows a network operator to deploy network functions as virtualized software instances instead of dedicated hardware appliances.
NFV MANO	Network Functions Virtualization Management and Organization
NFVI	NFV Infrastructure – entire NFV environment across all locations consisting of both hardware and software components.
NI-LR	Network Induced Location Request

Term or Acronym	Description
NIA	Non-IP Data Delivery (NIDD) Information Answer
NIC	Network Interface Card
NID	Network Identifier
NIDD	Non-IP Data Delivery
NIR	Non-IP Data Delivery (NIDD) Information Request
NMIS	Network Management Interface Specification
NOR	Notify Request
northbound	Interfaces from function under discussion towards the enterprise/application servers.
NPLI	Network Provided Location Information
NPU	Network Processing Unit
NRI	Network Resource Identifier
NS layer	Network Services layer – the transport protocol used between the SGSN and the BSS, specifically the PCUSN of the BSS, to ensure in-order delivery of packets sent between higher layer protocols.
NS-VC	Network Services Virtual Connection – an end-to-end logical connection for the transport of signaling and data packets between the SGSN and the BSS.
NSE	Network Services Entity – a logical entity that exists on both the SGSN and the BSS for managing the transport of both signaling and data packets between them.
NSSA	Not-So-Stubby-Area
NSWO	Non-Seamless WLAN Offload
NTP	Network Time Protocol
ntpd	Network Time Protocol daemon
NUMA	Non Uniform Memory Access
NVOD	Near Video on Demand
OA&M OAM	Operations, Administration and Maintenance – all the tasks necessary for providing, maintaining, or modifying the services provided by a switching system.
OC	Overload Control
OCI	Overload Control Information
OCS	Online Charging System
OCX	Operator Carrier Exchange
ODB	Operator Determined Barring
OEM	Original or Other Equipment Manufacturer

Term or Acronym	Description
OFCS	Offline Charging System
OI	Operator Identifier
OID	Object Identifier
OLR	Overload Control
OM	Operational Measurement
OMA	Open Mobile Alliance
OOM	Out Of Memory
OOS	Out Of Service
Opex	Operational Expenditure
OS	Operating System
OSGi	Open Services Gateway Initiative
OSS/BSS	Operational Support System/Business Support System
OSU	Online Software Upgrade
OTDOA	Observed Time Difference Of Arrival
OTT	Over-the-top – online delivery of video and audio without the Internet service provider’s involvement in the control or distribution of the content itself. Refers to content that arrives from a third party to the end user device, leaving the service provider responsible only for transporting IP packets.
OVA	Open Virtualization Archive
OVS	Open vSwitch – an open-source implementation of a distributed virtual multilayer switch providing a switching stack for hardware virtualization environments that supports multiple protocols/standards used in computer networks.
P-CSCF	Proxy Call Session Control Function
P-NIC	Physical Network Interface Controller
P-TMSI	Packet-Temporary Mobile Subscriber Identity
P2PTV	Peer-to-peer Television
PAT	Port Address Translation
PCB	Printed Circuit Board
PCC	Policy and Charging Control
PCEF	Policy and Charging Enforcement Function – an element that enforces the rules.
PCI	Peripheral Component Interconnect
PCO	Protocol Configuration Option

Term or Acronym	Description
PCP	Priority Code Point
PCRF	Policy and Charging and Rules Function – an engine that manages the rules. Supports service data flow detection, policy enforcement, and flow-based charging.
PCU	Packet Control Unit
PCUSN	Packet Control Unit Support Node – a node in the BSS that interacts either directly or indirectly with all the BSS nodes, except the Transcoder Unit (TCU).
PDCP-SN	Packet Data Convergence Protocol-Sequence Number
PDHIR	Packet Data Header Information Report
PDN	Packet Data Network – a communications system designed to carry packet data. Public Data Network
PDP	Packet Data Protocol Policy Decision Point
PDP Context	Packet Data Protocol context – a packet data connection over which the UE and the network can exchange IP packets. The information sets held in the MS and the GSNs for a PDP address. It contains information such as PDP Type, QoS Profile, APN, and GGSN. Usage of these packet data connections is restricted to specific services, accessible via APNs.
PDU	Protocol Data Units – a data unit that is transferred among peer entities of a network and contains protocol information, such as control information and address information.
PEC	Product Engineering Code
PEM	Power Entry Module
PEP	Policy Enforcement Point
PF	Physical Function
PFC	Packet Flow Context
PFCP	Packet Forwarding Control Protocol
PFE	Packet Forwarding Engine
PFM	Packet Flow Management
PGW	Packet Data Network Gateway
P-GW PGW	PDN Gateway
PGW-C	Packet Data Network Gateway - Control Plane
PGW-U	Packet Data Network Gateway - User Plane
PHB	Per-Hop Behavior
PICMG	PCI Industrial Computer Manufacturers Group

Term or Acronym	Description
PKI	Public Key Infrastructure – a certificate-based mechanism for public key distribution.
PLFP	Partner Link Failure Propagation
PLM	Product Line Management
PLMN	Public Land Mobile Network – a GSM, GPRS, or UMTS public mobile communications network established and operated by an administration or its licensed operator(s), for the specific purpose of providing land mobile communication services to the public.
PLS	Pooling Licensing Server
PM	Performance Management
PMC	PCI Mezzanine Card
PMTU	Path Maximum Transmission Unit
POJO	Plain Old Java Object – indicates that the object is not special, but is an ordinary Java Object (in particular is not an EJB).
PCRF	Policy and Charging and Rules Function – an Engine that manages the rules, supports service data flow detection, policy enforcement, and flow-based charging.
PoP	Point of Presence
POST	Power-On Self-Test
PPP	Point-to-Point Protocol
PPS	Packets Per Second
PPV	Pay-Per-View
PS	Packet Switched
PSAP	Public Safety Answering Points
PSK	Pre-Shared Key – a form of IKE authentication.
PVI	Pre-emption Vulnerability Indicator
PWS	Public Warning System
PXE	Preboot eXecution Environment
QCI	QoS Class Identifier
QEMU	Quick Emulator
QoE	Quality of Experience
QoS	Quality of Service
QPI bus	QuickPath Interconnect is a point-to-point processor interconnect that replaced the front-side bus (FSB) in certain desktop platforms.

Term or Acronym	Description
QSFP+	Quad Small Form-factor Pluggable Plus
QUIC	Quick UDP Internet Connection
QVOD	Quasi Video on Demand
RA	Router Advertisement
RAB	Radio Access Bearer
RAC	Routing Area Code
RADIUS	Remote Authentication (or Access) Dial-In User Service
RAI	Routing Area Identity
RAI-FQDN	Routing Area Identity-Fully Qualified Domain Name
RAID	Redundant Array of Independent Disks
RAM	Random Access Memory
RAN	Radio Access Network – the part of the mobile telecommunication system located between the mobile phone and the core network.
RANAP	Radio Access Network Application Part
RAR	Re-Auth-Request
RAT	Radio Access Technology Radio Access Type
RAU	Routing Area Update
RE	Routing Engine
REST	Representational State Transfer
REST/TCP	Representational State Transfer Interface between SCEF and SCS/AS
RFSP	Radio Access Technology Frequency Selection Priority
RGMII	Reduced Gigabit Media-Independent Interface
RHEL	Red Hat Enterprise Linux
RHOSP	Red Hat OpenStack Platform
RIA	Reporting Information Answer
RIM	Records and Information Management
RIP	Routing Information Protocol
RIR	Reporting Information Request
RLC	Radio Link Control

Term or Acronym	Description
RM	Resource Manager
RM VM	Resource Manager Virtual Machine
RMG	Routing Monitoring Group
RNC	Radio Network Controller
RoHS	Restriction of Certain Hazardous Substances Restriction of Hazardous Substances Directive
ROM	Read-Only Memory
RPC	Reduced Partial CDR
RPS	Receive Packet Steering
RRC	Radio Resource Control
RS	Router Solicitation
RSS	Receive Side Scaling
RSU	Requested Service Unit
RSZ	Regional Subscription Zone
RTM	Rear Transition Module
RTM	Rear Termination Module
RTMP	Real Time Messaging Protocol
RTP	Real Time Transport Protocol
RTSP	Real Time Streaming Protocol
RTT	Real-Time Troubleshooting
S-CDR	Serving GPRS Support Node - CDR
S-CSCF	Serving Call Session Control Function
SGW S-GW	Serving Gateway
SGW-C	Serving Gateway - Control Plane
SGW-U	Serving Gateway - User Plane
S-NAPTR	Straightforward-Naming Authority Pointer – limits the NAPTR record types to “a”, “s” and “”. S-NAPTR also simplifies the rewrite rules to replacement rules only.
S-SMO-CDR	SGSN delivered Short message Mobile Originated – CDR
S-SMT-CDR	SGSN delivered Short message Mobile Terminated – CDR

Term or Acronym	Description
S1-AP	S1 Application Protocol
S2b	Interface between the ePDG and PGW (GTPv2) used to transport WLAN UE data from non-3GPP untrusted access networks through IPsec and GTP tunnels for secure data transport and connectivity to the PDN.
S6b	Diameter-based interface between the 3GPP AAA Server and PGW used to manage address information, retrieve and update mobility-related information, authenticate APNs, and allocate UE IP addresses.
S6m	Interface between SCEF and HSS that supports subscriber information retrieval used to support device triggering.
S6t	Interface between SCEF and HSS
SA	Security Association – a unidirectional security agreement between two peers.
SA	System Architecture
SaaS	Software as a Service – a software delivery model in which software and its associated data are hosted centrally and are typically accessed by using a thin client, such as a Web browser over the Internet.
SAD	Security Association Database – A set of active Security Associations.
SAE	System Architecture Evolution
SAE-GW	Serving Gateway
SAF	Service Availability Forum
SAI	Service Area Identity
SaMOG	S2a Mobility based on GTP
SAS	SGSN Accounting Server – SGSN entity that provides SGSN billing functionality.
SATA	Serial Advanced Technology Attachment – a high-speed serial interface standard for storage devices.
SC	Subscriber Control: Primary purpose is to organize the related C-plane processes, such as GMM, SM, HLR Cache, and MAP Client.
SCCP	Signaling Connection Control Part
SCEF	Service Capability Exposure Function
SCF	Service Control Function
SCG	Secondary Cell Group
SCP	Service Control Point – an external entity used to manage Prepaid SMS subscriber accounts.
SCS	Service Capability Server
SCTP	Stream Control Transmission Protocol
SD	Subscriber Data – consists of the application handling the protocols involved in the 2G and 3G subscriber data path, namely LLC, SMDCP and GTP.

Term or Acronym	Description
SDC	Service Data Container
SDF	Service Data Flow
SDN	Software-defined networking – an approach to building data networking equipment by decoupling the control plane from the data plane.
SDR	Sensor Data Record
SDRAM	Synchronous Dynamic Random Access Memory
SECM	Security Module
SELV	Safety Extra Low Voltage Circuits
SeNB	Secondary eNB
SEND	Secure Neighbor Discovery protocol
SERDES	Serializer/Deserializer
Service Component	A service model is defined using service components. A service component consists of a set of vendor objects across one or more vendors and their associations with other vendor objects for provisioning or deprovisioning a component, such as a packet filter, service, rule, etc. Service components can be designed to access multiple device instances using different API protocols. See also Association Service Component (page 3) .
Service Component Instance	An actual instance of a service component running in a lab or production network and derived from a service component.
Service Instance	An actual instance of a service model running in a lab or production network. A service instance is derived from a service model and applied and rolled out to specific device instances in vTransactor.
Service Model	Represents a service that consists of service components and their associations. The service model is a template used to create instances of the “service” which is then applied and rolled out to specific vendor object instances.
SFP	Single Form-Factor Pluggable Transceiver
SFP	Small Form-Factor Pluggable
SFTP	Secure FTP
SGMII	Serial Gigabit Media Independent Interface
SGs	3GPP Interface between MME and MSC/VLR.
SGSN	Serving GPRS Support Node – main functions are to detect new GPRS mobile stations in its service area, send/receive data packets to/from the mobile stations, and record the location of mobile stations inside its service area.
SGW	Serving Gateway
SH-BGP	Single Hop Border Gateway Protocol
SIA	Subscriber Information Answer

Term or Acronym	Description
SIG	Signaling SS7/IP Gateway
SIG VM	Signaling Virtual Machine – a type of VM on the MCC MME.
SIGTRAN	Signaling Transport – defines a family of protocols that provide reliable datagram transport service and user adaptation layer for traditional Signaling System 7-based protocols.
SIM	Subscriber Identity Module
SIR	Subscriber Information Request
SLB	Steering Load Balancer – a type of VM for the MCC MME. Session Load Balancer
SLF	Subscriber Locator Function
SLg	Diameter interface between the MME and GMLC.
SMPP	Short Message Peer-to-Peer Protocol
SMS	Short Messages Service – provides the means to transfer messages between a GPRS PLMN MS and a Short Message entity through a service center.
SMSC	Short Message Service Center
SMTP	Simple Mail Transfer Protocol
Snapshot	A version of a service instance in vTransactor. When you make changes and updates to a specified service instance and then click the Snapshot icon, vTransactor creates a new snapshot of that service instance. If you make one additional change and do not click the Snapshot icon, the changes will be pending and will not be part of the latest snapshot (until you click the Snapshot icon). Snapshots can be compared with devices (Config Diff) and can be rolled out to a selected Footprint. See also Footprint (page 11) .
SNDCP	Sub Network Dependent Convergence Protocol – a layer 3 protocol responsible for segmentation or reassembly and possibly compression of data packets for transport between the MS and the SGSN.
SNI	Server Name Indication
SNMP	Simple Network Management Protocol – used by network management systems to monitor network-attached devices for conditions requiring administrative attention.
SNOW3G	A stream cipher that uses a 128-bit key and IV.
SNR	Seamless National Roaming
SNS	Sub-Network Services
SOAP	Simple Object Access Protocol – protocol used for exchanging XML-based messages over computer network, normally using HTTP/HTTPS. SOAP forms the foundation layers of the Web Services stack, providing a basic messaging framework.
SON	Self Optimized Networks
southbound	Interfaces from function under discussion towards the 3GPP network functions.

Term or Acronym	Description
SPA	Security Policy Agent
SPD	Security Policy Database Rule set used by an IPsec implementation to secure traffic.
SPI	Security Parameter Index Serial Peripheral Interface
SPM	Security Policy Manager
SPR	Subscriber Profile Repository
SR-IOV	Single Root I/O Virtualization
SRAM	Static Random Access Memory
SRNS	Serving Radio Network Subsystem
SRV	Service records – a specification of data in the Domain Name System defining the location (hostname and port number) of servers for specified services defined in RFC 2782.
SS7	Signaling System 7 – common channel signaling system designed for the control of voice and non-voice services and for use in a digital environment in which signaling information may be sent at 64 kbit/s.
SSD	Solid State Drive – an all-electronic storage device that is an alternative to a hard disk drive. Used in smartphones, MP3 players, digital cameras and laptop computers, solid state drives are faster than hard disks because there is zero latency (no read/write head to move).
SSH	Secure Shell
SSM	Subscriber Services Module – a cryptographic network protocol for operating network services securely over an unsecured network.
SSO	Single Sign-on – provides single authentication/sign-on for multiple applications.
STA	Session-Termination-Answer Diameter command code
STa	Interface between the non-3GPP access network and the 3GPP AAA server or between a non-3GPP access network and the 3GPP AAA proxy.
STI	Standard Transcoding Interface
STP	Shielded Twisted Pair
STPx	System Test Phase <x> (milestone)
STR	Session Termination Request
SU	Service Unit
SVOD	Subscription Video on Demand
SWm	Interface linking the 3GPP AAA server and the ePDG or between the 3GPP AAA Proxy and the ePDG.

Term or Acronym	Description
SWu	Interface between the ePDG and 3GPP Diameter-based AAA Server used to authenticate UE through EAP and IKEv2 signaling, secure data transport through IPsec tunnels, and IP address allocation to UEs. This interface is from the ePDG to UE in non-3GPP untrusted IP access networks. It supports roaming/mobility, tunnel authentication and authorization of subscribers.
SX	Short Range Optical Transceiver
T-PDU	Tunneled PDU
T4	Interface between SCEF and SMS-SC that supports the transfer of device trigger messages and the success/failure reporting of device trigger deliveries to the UE.
T6a	Interface between SCEF and MME
T8	Interface between SCEF and SCS/AS
TA	Tracking Area
TAC	Tracking Area Code
TACACS+	Terminal Access Controller Access-Control System Plus
TAI	Tracking Area Identity
TAM	Traffic Anchoring Mode
TAS	Technical Assistance Services
TAU	Tracking Area Update
TBD	To Be Determined
TCAP	Transaction Capabilities Application Part – TCAP uses the MTP and SCCP software layers to provide connectionless signaling for transaction services. Simply put, it allows two nodes to communicate with each other in the same language.
TCE	Trace Collection Entity
TCP	Transmission Control Protocol
TCP/IP	Transmission Control Protocol/Internet protocol – a protocol stack - designed to connect different networks - on which the Internet is based.
TDF	Traffic Detection Function
TDM	Time Division Multiplex
TDP	Trigger Detection Point
TEID	Tunnel End Identifier
TFD	Toll Free Data
TFT	Traffic Flow Template
TI	Technology Introduction

Term or Acronym	Description
TIPC	Transparent Inter Process Communication
TLLI	Temporary Logical Link Identifier – describes an identified MS registered in the network and owning a LLC context with the SGSN.
TLS	Transport Layer Security
TLV	Type Length Value
TODA	Time Of Day Accounting
TOS	Type of Service Terms of Service
TPE	Twisted-Pair Ethernet
TPS	Transactions Per Second
TTCT	Tariff Time Change Trigger
TUI	text user interface
TVOD	Transactional Video on Demand
TWAG	Trusted WLAN Access Gateway
TWAP	Trusted WLAN AAA Proxy
U-Plane	User Plane – a function associated with the transfer of actual user data to and from a PDN.
UAProf	User Agent Profile
UART	Universal Asynchronous Receiver / Transmitter
UBR	Unspecified Bit Rate
UDP	User Datagram Protocol
UDP/IP	User Datagram Protocol/Internet Protocol – a transport layer, connectionless mode protocol, providing a datagram mode of communication for delivery of packets to a remote or local user.
UE	User Equipment – a device used directly by an end-user to communicate in UMTS and LTE.
UE-AMBR	User Equipment – Aggregated Maximum Bit Rate
UGL	Update GPRS Location
UIM	User Identity Module
UL	Underwriters Laboratories
ULI	User Location Information
UMTS	Universal Mobile Telecommunications System
UP	User Plane – see U-Plane.

Term or Acronym	Description
UPF	User Plane Function
UPM	User Datagram Protocol (UDP) Path Manager – a software component on the MME.
UPSM	User Datagram Protocol (UDP) Path Steering Manager
URI	Universal Resource Indicator
USD	UMTS Subscriber Data
USIM	UMTS SIM
USU	Used Service Units
UTC	Universal Time Coordinated
UTRA	Universal Terrestrial Radio Access
UTRAN	Universal Terrestrial Radio Access Network
VAS	Value-Added Service
VASP	Value-Added Service Provider
VBR	Variable Bit Rate – the bit rate used in sound or video encoding. VBR files vary the amount of output data per time segment. VBR allows a higher bit rate (and therefore more storage space) to be allocated to the more complex segments of media files while less space is allocated to less complex segments.
vBuilder	An ASAP tool that works in a carrier's corporate network to construct service orchestration templates or <i>recipes</i> that perform management operations on any class of network device or application using a connector type such as YANG or REST.
VBMC	Virtual Baseboard Management Controller
vCPU	Virtual Central Processing Unit is assigned to each VM in a cloud environment.
vDISK (or VD)	Virtual Disk – a virtual device that provides an area of usable storage capacity on one or more physical disk drive(s) in a computer system.
Vendor Component	A vendor component represents a collection of vendor objects to be applied on a device type. A vendor component is associated with only one device type and one API protocol type. A service component may have several vendor components.
Vendor Object	A device type may have one or more vendor objects used in a service model and based on an API protocol or CLI. Typically, vendor objects are parsed from the vendor schema derived from the NETCONF/YANG or REST API. A subject matter expert (SME) may also create the vendor objects manually, for example, for CLI (SSH). A SME may configure a specific vendor object of a device type with parameters or hard code the values of the parameters.
Vendor Object Instance	One or more instances of a vendor object contained by any number of service instances.
Vendor Schema Object	Refers to a supported API protocol (e.g., YANG or REST) to import and on which to base and create vendor objects in vBuilder.
vEPC	virtualized Evolved Packet Core – a mobile core network system that accommodates LTE access systems.

Term or Acronym	Description
vePDG	Virtualized Evolved Packet Data Gateway
VIM	Virtualized Infrastructure Manager
VLAN	Virtual Local Area Network
VLR	Visitor Location Register
VM	Virtual Machine
vMCM	Virtual Management Configuration Module – a VM instance that runs core processes, including node and configuration functions.
vMemory	Virtual Memory – a container for managing an array of memory containers.
VMM	Virtual Machine Management
VNF	Virtual Network Function – software implementation of a network function that can be deployed on an NFVI.
VNFM	Virtual Network Function Manager
vNIC	Virtualized Network Interface Card – a vNIC is a VM that provides virtual network interface by defining a range of MAC addresses.
VOD	Video On Demand
VPLMN	Visited Public Land Mobile Network
VPN	Virtual Private Network
VRF	Virtual Routing and Forwarding
VSA	Vendor-Specific Attribute
vSSF	Virtual Slice Selection Function
vSwitch	Virtual Switch – a software application that allows communication between VMs.
vTransactor	An ASAP tool installed in customer NOCs to implement the service orchestration recipes in a carrier's network.
WAE	Wireless Application Environment
WAG	Wireless Access Gateway
WAP	Wireless Application Protocol
WCDMA	Wideband Code Division Multiple Access
WDP	Wireless Datagram Protocol
WEEE	Waste from Electrical and Electronic Equipment
WLAN	Wireless Local Area Network
WP-HTTP	Wireless Profiled HTTP

Term or Acronym	Description
WP-TCP	Wireless Profiled TCP
WSM	Workflow Services Module – a VM instance that provides workflow services.
WSP	Wireless Session Protocol
WTLS	Wireless Transport Layer Security
WTP	Wireless Transaction Protocol
X.509	A specific implementation of PKI using trusted certificates.
XML	Extensible Markup Language